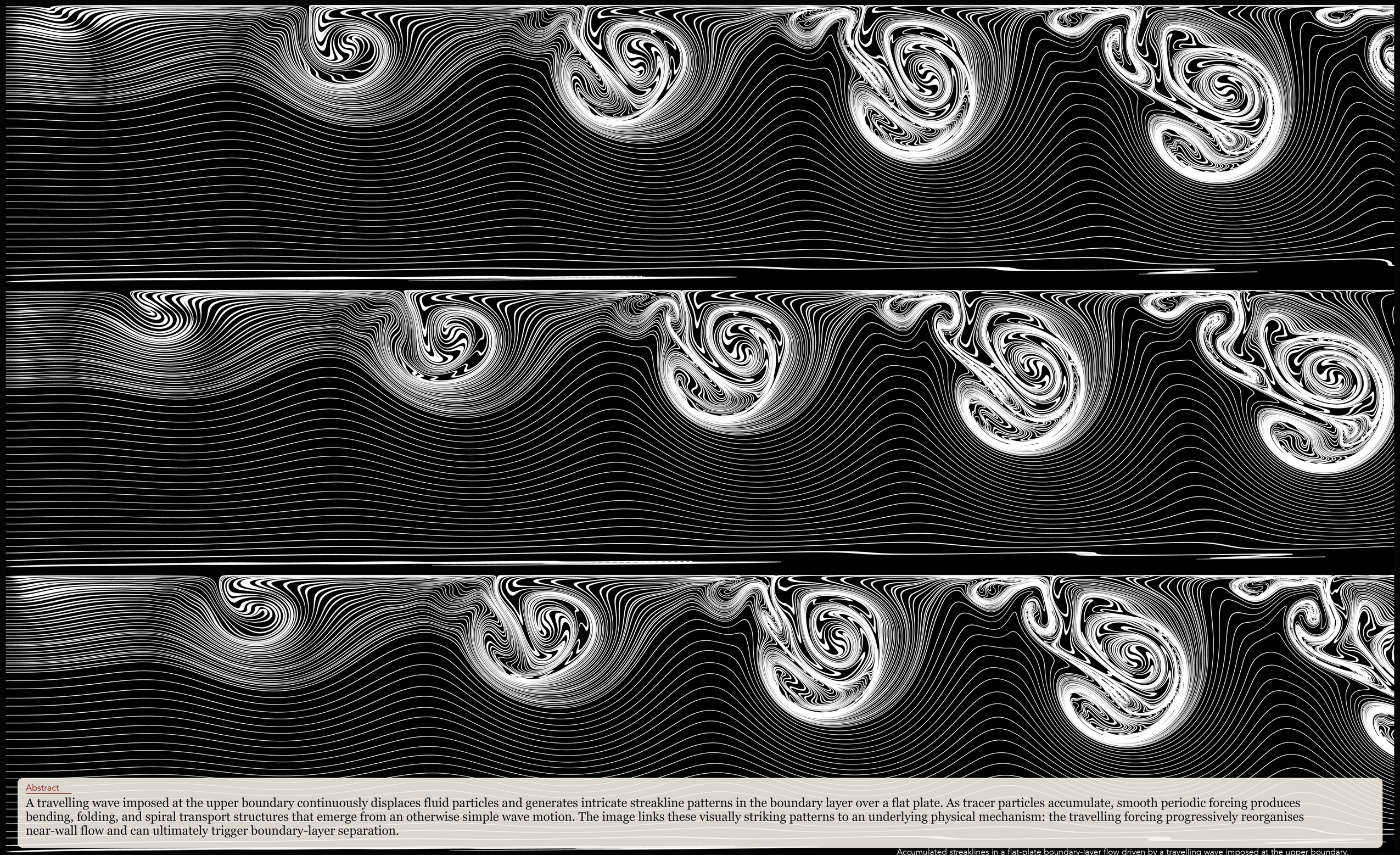
# From Waves to Separation: Streaklines Generated by Travelling-Wave Forcing

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Abstract

A travelling wave imposed at the upper boundary continuously displaces fluid particles and generates intricate streakline patterns in the boundary layer over a flat plate. As tracer particles accumulate, smooth periodic forcing produces bending, folding, and spiral transport structures that emerge from an otherwise simple wave motion. The image links these visually striking patterns to an underlying physical mechanism: the travelling forcing progressively reorganises near-wall flow and can ultimately trigger boundary-layer separation.

Accumulated streaklines in a flat-plate boundary-layer flow driven by a travelling wave imposed at the upper boundary.